

Publications

Books

- [1] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, *Design Automation for Field-coupled Nanotechnologies*. Springer, 2022. DOI: 10.1007/978-3-030-89952-3

Journal Articles

- [2] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, “RTL-to-Atoms Synthesis of a Machine Learning Accelerator on Atomic-Scale Computers,” *IEEE Transactions on Nanotechnology (TNANO)*, 2026. DOI: 10.1109/TNANO.2026.3689032
- [3] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “QuickCell: Fast Automatic Design of Standard Cells for Silicon Dangling Bond Logic,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3605537
- [4] S. Hofmann, **M. Walter**, R. Wille, “Efficient and Scalable Post-Layout Optimization for Field-coupled Nanotechnologies,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3549354
- [5] S. Hofmann, **M. Walter**, R. Wille, “Graph-Oriented Layout Design for Field-coupled Nanocomputing via Parallel Multi-Objective Search Space Exploration,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2025. DOI: 10.1109/TCSI.2025.3614709
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- [8] F. Sill Torres, P. A. Silva, G. Fontes, **M. Walter**, J. A. M. Nacif, R. S. Ferreira, O. P. V. Neto, J. F. Chaves, R. Wille, P. Niemann, D. Große, R. Drechsler, “On the Impact of the Synchronization Constraint and Interconnections in Quantum-dot Cellular Automata,” *Microprocessors and Microsystems (MICPRO)*, 2020.
- [9] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, “Placement and Routing for Tile-based Field-coupled Nanocomputing Circuits is NP-complete (Research Note),” *Journal on Emerging Technologies in Computing Systems (JETC)*, 2019. DOI: 10.1145/3312661

Conference Proceedings

- [10] J. Drewniok, **M. Walter**, R. Wille, “Operational Domain Sketching for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [11] M. Feldmeier, **M. Walter**, R. Wille, “A Framework for Post-Mapping Feedback-Guided Selective Logic Transformations,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [12] I. V. Gardner, **M. Walter**, R. Wille, M. Cochez, “AIG2PT: A Generative Pre-trained Transformer for Unconditional And-Inverter Graph Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [13] B. Hien, **M. Walter**, R. Wille, “Minimizing Area and Delay in Planar Physical Design for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [14] B. Hien, **M. Walter**, R. Wille, “Toward Compact Planar Circuits for Atomic-Scale Computing: A Hybrid Crossing Elimination Flow,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [15] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Holistic Physical Design for Silicon Atomic Logic: Cramming More Components onto Clock Zones,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [16] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Mastering the Exponential Complexity of Exact Physical Simulation of Silicon Dangling Bonds,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026.

- [17] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, "Design and Emulation Methodology for Atomic-Scale Systolic Arrays: An LLM Accelerator Case Study in Silicon DB Logic," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
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- [19] **M. Walter**, M. Feldmeier, R. Wille, "Exact Synthesis with Optimal Switching Activity," in *Design, Automation and Test in Europe (DATE)*, 2026.
- [20] **M. Walter**, I. V. Gardner, R. Wille, "aigverse: A Unified Infrastructure for Machine Learning-Driven Logic Synthesis," in *International Workshop on Logic Synthesis (IWLS)*, 2026.
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- [27] S. Hofmann, J. Drewniok, **M. Walter**, R. Wille, "MNT Designer: A Comprehensive Design Tool for Field-coupled Nanocomputing," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025.
- [28] S. Hofmann, **M. Walter**, R. Wille, "Late Breaking Results: Physical Co-Design for Field-coupled Nanocomputing," in *Design, Automation and Test in Europe (DATE)*, 2025.
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- [35] J. Drewniok, **M. Walter**, R. Wille, "The Need for Speed: Efficient Exact Simulation of Silicon Dangling Bond Logic," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. DOI: 10.1109/ASP-DAC58780.2024.10473946
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